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KOMSAS GO: AN AUGMENTED REALITY LEARNING MODULE MALAY LANGUAGE FOR UPPER FORM STUDENT

GROUP MEMBERS

Name	Matric No
MUHAMMAD NAJMI HAFIY BIN MOHMAD NAZER	17DIT24F1112
MUHAMMAD HAFIZUDDIN BIN SHAHRUN NIZAL	17DIT24F1092
MUHAMMAD ADAM HAFIY BIN ZAHARIMAN	17DIT24F1003

SUPERVISOR: TS. KHALID BIN DERIS

SESSION I: 2026/2027

DECLARATION

We hereby declare that the technical report entitled KOMSAS GO: AN AUGMENTED REALITY LEARNING MODULE MALAY LANGUAGE FOR UPPER FORM STUDENT is our own work, prepared with guidance from TS. KHALID BIN DERIS. All sources, references, and quotations used in this report have been properly acknowledged. We also confirm that this report has not been submitted previously or concurrently for any other diploma, qualification, or award at any institution.

1. Signature : 

   Name : MUHAMMAD NAJMI HAFIY BIN MOHMAD NAZER

   Registration No : 17DIT24F1112

   Date : 8/10/2026

2. Signature : 

   Name : MUHAMMAD HAFIZUDDIN BIN SHAHRUN NIZAL

   Registration No : 17DIT24F1092

   Date : 8/10/2026

3. Signature : 

   Name : MUHAMMAD ADAM HAFIY BIN ZAHARIMAN

   Registration No : 17DIT24F1003

   Date : 8/10/2026

APPROVAL FOR SUBMISSION

Project Title: KOMSAS GO: AN AUGMENTED REALITY LEARNING MODULE MALAY LANGUAGE FOR UPPER FORM STUDENT

Session: I 2026/2027

Submitted by:

1. MUHAMMAD NAJMI HAFIY BIN MOHMAD NAZER (17DIT24F1112)
2. MUHAMMAD HAFIZUDDIN BIN SHAHRUN NIZAL (17DIT24F1092)
3. MUHAMMAD ADAM HAFIY BIN ZAHARIMAN (17DIT24F1003)

"This technical report has been read, checked and approved in terms of scope and quality for the award of Diploma in Information Technology."

Checked by

Supervisor's Signature: 

Supervisor's Name: TS. KHALID BIN DERIS

Date: 8/10/2026

ABSTRACT

KOMSAS GO is an innovative Augmented Reality (AR) mobile application designed to revitalize the Malay Literature (KOMSAS) component within the upper secondary Malay Language curriculum. While KOMSAS is vital for fostering literary appreciation and cultural heritage, conventional text-heavy methods often lead to passive learning and student disinterest. This project addresses these challenges by integrating modern AR technology to create an interactive, visually rich educational experience. Developed under the ADDIE instructional design framework, KOMSAS GO is built using the Unity game engine and Vuforia AR SDK. The application operates by scanning physical textbook pages or custom AR flashcards to trigger dynamic digital content. Core literary elements such as plotlines, character profiles, central themes, and moral values are rendered into interactive 3D models, character animations, localized voiceovers, and gamified assessments. Usability testing and effectiveness evaluations involving students and teachers indicate a marked increase in motivation and superior retention of complex literary concepts. The module enables self-directed learning, allowing students to grasp abstract literary structures far more intuitively than through printed text alone. Ultimately, KOMSAS GO demonstrates the successful integration of emerging technology within the Malaysian secondary school system. By seamlessly connecting physical print media with digital interactivity, the application serves as a scalable model for 21st-century learning (PAK-21), contributing to the modernization of Malay Language education.

Keywords: Augmented Reality (AR), Malay literature education, Educational Application

ABSTRAK

KOMSAS GO merupakan sebuah aplikasi mudah alih Realiti Terimbuh (Augmented Reality, AR) yang inovatif dan dibangunkan bagi memperkasa pembelajaran komponen Kesusasteraan Melayu (KOMSAS) dalam kurikulum Bahasa Melayu peringkat menengah atas. Walaupun KOMSAS memainkan peranan penting dalam memupuk penghayatan sastera serta menghargai warisan budaya, kaedah pembelajaran konvensional yang terlalu bergantung pada teks sering menyebabkan pembelajaran menjadi pasif dan kurang menarik minat pelajar. Projek ini dibangunkan bagi menangani cabaran tersebut melalui pengintegrasian teknologi AR moden untuk menghasilkan pengalaman pembelajaran yang interaktif, menarik secara visual dan lebih bermakna. KOMSAS GO dibangunkan berdasarkan kerangka reka bentuk pengajaran ADDIE dengan menggunakan enjin permainan Unity dan Vuforia AR SDK. Aplikasi ini berfungsi dengan mengimbas halaman buku teks fizikal atau kad imbas AR khas untuk mengaktifkan kandungan digital secara dinamik. Elemen utama kesusasteraan seperti plot, watak dan perwatakan, tema, serta nilai dan pengajaran dipersembahkan melalui model 3D interaktif, animasi watak, rakaman suara dalam bahasa tempatan serta aktiviti penilaian berasaskan permainan. Ujian kebolehgunaan dan penilaian keberkesanan yang melibatkan pelajar serta guru menunjukkan peningkatan yang ketara dari segi motivasi pembelajaran dan keupayaan pelajar mengingati konsep sastera yang kompleks. Modul ini turut menyokong pembelajaran kendiri, sekali gus membolehkan pelajar memahami struktur dan elemen sastera yang bersifat abstrak dengan lebih mudah dan intuitif berbanding penggunaan bahan bercetak semata-mata. Secara keseluruhannya, KOMSAS GO membuktikan bahawa teknologi baharu dapat diintegrasikan secara berkesan dalam sistem pendidikan sekolah menengah di Malaysia. Dengan menghubungkan bahan bercetak fizikal dan kandungan digital interaktif secara lancar, aplikasi ini berpotensi menjadi model pembelajaran yang boleh diperluaskan selaras dengan konsep Pembelajaran Abad Ke-21 (PAK-21) serta menyumbang kepada pemodenan pengajaran dan pembelajaran Bahasa Melayu.

Kata Kunci: Realiti Terimbuh (AR), Pendidikan Kesusasteraan Melayu, Aplikasi Pendidikan

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1.0 PROJECT PLAN

1.1 INTRODUCTION

Digital technology has become an important element in modern education by providing more interactive and engaging learning experiences. In Malay Literature (KOMSAS) education, students often face difficulties in understanding literary elements such as characters, themes, values and messages. Traditional learning methods that mainly depend on textbooks and classroom explanations may limit students' ability to visualise literary content and reduce their interest in learning KOMSAS.

To overcome these challenges, the AR KOMSAS GO: Augmented Reality KOMSAS Learning Application was developed as a mobile-based educational platform for secondary school students. The application integrates Augmented Reality (AR) technology to allow students to interact with digital representations of KOMSAS characters and access related learning information through mobile devices.

Besides AR learning, the application includes interactive quizzes, Literature Anthology, Bahasa Istana resources and the KOMSAS AI Tutor to provide additional learning support. Gamification elements such as experience points (XP), rewards and badges are also implemented to encourage student participation and motivation throughout the learning process.

Furthermore, AR KOMSAS GO provides a Teacher Dashboard that enables teachers to monitor student progress, create quizzes, manage learning missions and provide rewards. By combining AR, gamification and Artificial Intelligence, this project aims to provide a more interactive and effective platform for KOMSAS learning while supporting teachers in managing digital learning activities.

CLIENT:

Table 1.1: Client Background

DETAIL	DESCRIPTION
NAME	SMK TAMAN KLANG UTAMA
LOCATION	Jalan Sungai Keramat 40, Taman Klang Utama, 42100 Klang, Selangor
CORE SERVICE	Education
PRINCIPAL	Pn Hajah Umi Kalsom Binti Mohd Wahi
ACADEMIC SENIOR ASSISTANT	Pn Jaya A/P Ramasamy
HEAD PANEL OF BAHASA MELAYU UNIT	En. Mohmad Nazer Bin Hasim
EMPLOYEES	117
STUDENTS	2156
CLIENT UNIT	Bahasa Melayu

Initially, an interview was conducted with Mr. Mohmad Nazer Bin Hasim, a Head Panel of Bahasa Melayu Unit at SMK Taman Klang Utama to identify the challenges faced in the teaching and learning of KOMSAS. Based on the discussion, several issues were identified:

i. Many students experience difficulties in understanding KOMSAS literary elements such as characters, themes, values and messages due to the abstract nature of literature content.

ii. Students show limited engagement and interest towards KOMSAS learning because conventional learning methods mainly depend on textbooks, notes and classroom explanations.

iii. Teachers face challenges in providing interactive learning activities and visual learning materials that can help students better understand literary works.

iv. Monitoring students' learning progress and participation in KOMSAS activities can be challenging when using traditional teaching approaches.

In addition to the interview, a questionnaire was distributed to students to gather their perspectives regarding the use of digital technology in KOMSAS learning. The findings from the survey revealed:

i. Most students agreed that KOMSAS learning can be improved through more interactive and visual learning approaches.

ii. Students indicated that understanding characters, story plots and literary elements were among the challenging aspects of learning KOMSAS.

iii. Students showed interest in using mobile applications that provide interactive content such as Augmented Reality (AR), quizzes and digital learning materials.

iv. The majority of students preferred learning methods that combine visual elements, interactive activities and technology-based approaches compared to traditional reading methods only.

These findings highlight the need for an interactive digital learning platform that can improve students' engagement, enhance understanding of KOMSAS content and support teachers in delivering a more effective learning experience. Therefore, AR KOMSAS GO was proposed as a solution by integrating Augmented Reality, gamification features, Literature resources, AI Tutor and Teacher Dashboard to support both students and teachers in the KOMSAS learning process.



Figure 1.1: Interview Verification from SMK Taman Klang Utama

1.2 PROBLEM STATEMENT

The problem statements for the project are as follows:

1. **Text-Centric and Static Learning Methods**

   The teaching and learning of KOMSAS still relies heavily on printed textbooks and conventional text-based materials. This approach provides limited visual and interactive elements, making it difficult for students to imagine literary settings, characters, and events. As students today are highly exposed to digital technology, static learning materials may reduce their engagement and interest in learning KOMSAS.

2. **Unengaging and Conventional Assessment Methods**

   KOMSAS assessment activities are commonly conducted through worksheets, written exercises, and paper-based questions. These methods may become repetitive and provide limited interaction or immediate feedback to students. As a result, students may feel less motivated to revise important literary elements such as plot, themes, characters, moral values, and lessons.

3. **Difficulty in Understanding Complex Literary Language**

   Students often face difficulties in understanding classical prose, traditional expressions, and royal court language used in KOMSAS texts. These unfamiliar language structures can make literary content difficult to interpret and may cause students to lose interest. Without additional visual or interactive support, students may struggle to understand the meaning, context, and important literary elements presented in the text.

1.3 OBJECTIVE

The objectives of the projects are as follows:

i. To design an AR-based learning environment featuring dynamic 3D settings and character maps to enhance student engagement.

ii. To develop an AR application with interactive glossaries and visual aids that translate classical terms into modern language.

iii. To replace static, text-based learning materials with multi-sensory Augmented Reality experiences that cater to visual, auditory, and kinesthetic learners.

1.4 SCOPE

User Scope

i. **Student**

- Log in and manage their learning profile.
- Scan KOMSAS characters using AR technology.
- View character information, audio narration and learning materials.
- Answer KOMSAS-related quizzes.
- View scores, rewards, badges and learning levels.
- Access the Literature Anthology and Bahasa Istana module.
- Use the KOMSAS AI Tutor to ask questions about KOMSAS.
- View their position on the leaderboard.
- Complete learning missions.

ii. **Teacher**

- Log in to the Teacher Dashboard.
- View student information and student lists.
- Monitor student learning performance.
- View quiz records and learning progress.
- Create quiz questions for students.
- Create and manage learning missions.
- Give rewards to students.
- Give rewards to the entire class.
- Monitor student achievements and learning activities.

System Scope

i. **AR Character Learning**

- Scan KOMSAS character markers using AR technology.
- Display 3D KOMSAS characters in an augmented environment.
- Display character information and learning content.
- Provide audio narration for selected characters.
- Allow students to access character-related learning materials.

ii. **KOMSAS Quiz**

- Provide KOMSAS-related quiz questions.
- Allow students to submit answers.
- Calculate and display quiz scores.
- Provide feedback based on quiz performance.
- Record quiz progress for student learning records.

iii. **Gamification**

- Award XP and coins based on learning activities.
- Provide badges for achieving specific learning milestones.
- Display student levels and learning progress.
- Provide learning missions for students to complete.
- Display rewards and achievements.

iv. **Literature Learning**

- Provide access to the Literature Anthology.
- Provide poetry and traditional prose learning materials.
- Provide Bahasa Istana terminology and meanings.
- Allow students to explore literature content through the application.

v. **KOMSAS AI Tutor**

- Allow students to ask questions related to KOMSAS.
- Provide AI-generated explanations in Bahasa Melayu.
- Support questions about themes, characters, values and literary concepts.
- Provide immediate learning assistance to students.
- Restrict the tutor's responses to educational KOMSAS-related topics.

vi. **Student Progress and Leaderboard**

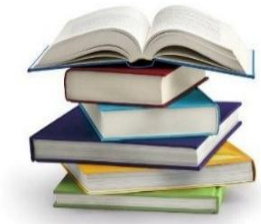
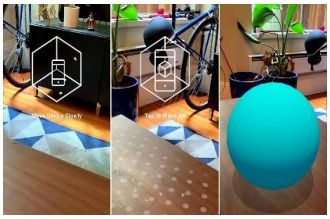
- Display student learning progress and achievements.
- Display student levels, XP and collected badges.
- Provide school and friend leaderboard rankings.
- Allow students to monitor their learning performance.

vii. **Teacher Management**

- Allow teachers to view student information.
- Allow teachers to monitor student learning performance.
- Allow teachers to create quiz questions.
- Allow teachers to create and manage learning missions.
- Allow teachers to provide rewards to individual students or the whole class.
- Display student quiz and learning records.

1.5 LITERATURE REVIEW

Table 1.2: Comparison of Existing Learning Approaches and AR-Based Systems

System Name	Traditional Textbook Methods 	Global Crowdsourced Apps 	Enterprise AR Frameworks (Unity + AR Foundation) 
Platform	Physical Print Media	Web & Mobile App	Cross-Platform Deployment (iOS / Android)
Description	Standard textual books used by teachers to deliver regional literature curriculum.	General learning or reference modules that host crowd-contributed global text data.	Professional development pipelines that build robust 3D/AR overlays and applications.
Disadvantages	Heavy reliance on rote memorization, lacks interactive components, and fails to cater to diverse learning styles.	Focuses on global educational content rather than specific local MOE curriculum alignments or detailed KOMSAS texts.	Can become overly complex to structure without localized asset mapping, demanding high initial device processing thresholds.

Several existing methods and digital frameworks were evaluated to support the creation of the interactive Augmented Reality learning module for upper form students. Among them:

- **Traditional Textbook Methods** offer static, paper-driven frameworks for analysing text structures. While they secure the core curriculum, their physical medium lacks dynamic visuals and instant audio feedback, leading to lower engagement among visual and auditory learners.

- **Global Educational Literature Frameworks** provide extensive digital databases for general reading and research. Although useful for broad literacy, they lack local relevance as they do not reflect the specific cultural themes, history, or linguistic elements of Malaysian KOMSAS literature.

- **Enterprise AR Toolsets (Unity, ARCore, ARKit)** deliver advanced cloud deployment, cross-platform engines, and robust camera tracking. However, they require complex technical expertise and high-end hardware, limiting accessibility on mid-range student mobile devices.

KOMSAS GO delivers a superior digital transformation by being curriculum-aligned, user-friendly, and affordable. It brings prescribed literature to life via 3D visualization, offering an immersive learning experience while providing teachers with a structured framework to track progress.

1.6 METHODOLOGY

Agile methodology was used to develop AR KOMSAS GO through an iterative and incremental development process. The methodology was selected because it allows the system to be developed in small iterations, tested regularly and improved based on feedback and identified issues. Each iteration focused on developing and refining specific system features such as Augmented Reality character scanning, KOMSAS quizzes, gamification, Literature Anthology, Teacher Dashboard and the KOMSAS AI Tutor.

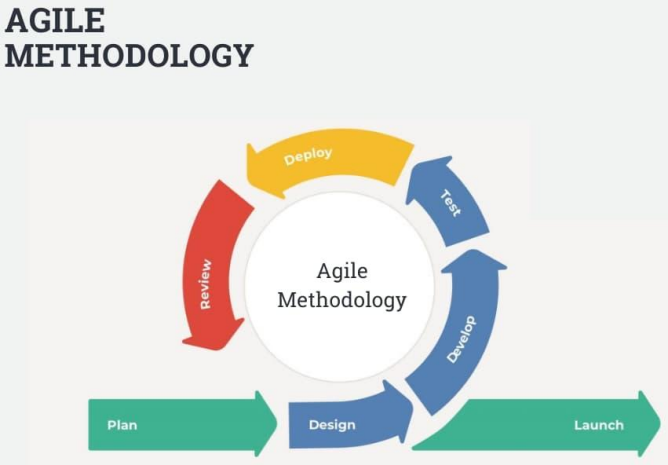


Figure 1.2: Agile Methodology

The benefits of using Agile methodology for this project are as follows:

i. Iterative Development

Each module (VR simulation, survival skills, signaling techniques) can be developed and improved in short cycles, allowing early detection of issues.

ii. Continuous Feedback and Testing

Regular testing throughout development ensures each feature works correctly before moving to the next stage, improving overall system quality.

iii. Flexibility to Changes

Agile allows the team to adjust features or fix issues quickly without disrupting the overall project timeline.

iv. Faster Delivery of Working Features

Since features are built incrementally, functional parts of the system can be demonstrated and evaluated early.

Table 1.3: Phase Agile Methodology

No	Phase	Activities
1	Planning	• Confirm project title and validate the project proposal. • Conduct literature review on Augmented Reality, gamification, Artificial Intelligence and digital learning applications. • Identify challenges in KOMSAS learning through interview and requirement gathering with SMK Taman Klang Utama. • Gather system requirements and define the project scope.
2	Design	• Design the overall system architecture of AR KOMSAS GO. • Design the user interface (UI) and user interaction flow for student and teacher modules. • Design the AR learning flow including character marker scanning and information display. • Design the database structure for student progress, quizzes, rewards and badges. • Plan the gamification mechanism, AI Tutor workflow and Teacher Dashboard features.

3	Development	• Set up the Unity project and configure AR development environment using Vuforia SDK. • Develop the student module including AR Learning, KOMSAS Quiz and Literature Anthology features. • Develop the KOMSAS AI Tutor for providing KOMSAS-related learning assistance. • Develop gamification features including XP, rewards and badges. • Develop the Teacher Dashboard for monitoring student progress and managing learning activities. • Integrate Firebase database for storing user data, progress and system information. • Create and integrate 3D KOMSAS character models and learning content. • Optimize application performance and improve user experience.
4	Testing	• Conduct unit testing for individual application functions. • Perform integration testing between AR module, quiz system, gamification system and database. • Test AI Tutor response functionality and user interaction. • Verify student progress tracking, rewards and badge systems. • Identify and fix system errors to improve application stability.
5	Deployment	• Conduct functional testing of AR KOMSAS GO application on Android devices. • Perform user acceptance testing with students and teachers. • Collect feedback regarding usability, interaction and learning experience. • Fix bugs and optimize the final application performance. • Prepare final APK build for project demonstration.
6	Review	• Finalize project report and documentation. • Prepare presentation slides and project demonstration materials. • Perform final verification of all system features. • Submit the completed project and conduct final presentation evaluation.

1.7 GANTT CHART

PROJECT NAME	AR KOMSAS GO: AN AUGMENTED REALITY LEARNING MODULE MALAY LANGUAGE FOR UPPER FORM STUDENT	START DATE	END DATE	OVERALL PROGRESS	MEMBER: MUHAMMAD ADAM HAFIY BIN ZAHARIMAN
PROJECT MANAGER	MUHAMMAD NAJMI HAFIY BIN MOHMAD NAZER	19/1/2026	21/10/2026	COMPLETED	MEMBER: MUHAMMAD HAFIZUDDIN BIN SHAHRUN NIZAL

Table 1.4: Project's Gantt Chart

WBS	TASK NAME	PLAN DURATION	START	FINISH
0.0	**AR KOMSAS GO: Augmented Reality KOMSAS Learning Application**	**176 Days**	**19/1/2026**	**21/10/2026**
1.0	**PHASE 1: PLANNING**	**39 Days**	**19/1/2026**	**26/2/2026**
1.1	Topic confirmation and proposal validation	5 Days	19/1/2026	23/1/2026
1.2	Literature review on AR, gamification and digital learning	11 Days	2/2/2026	12/2/2026
1.3	Requirement gathering through interview and questionnaire (SMK Taman Klang Utama)	11 Days	16/2/2026	26/2/2026
1.4	System requirements and project scope finalization	12 Days	27/2/2026	12/3/2026
2.0	**PHASE 2: DESIGN**	**42 Days**	**13/3/2026**	**23/4/2026**
2.1	Overall system architecture design	6 Days	13/3/2026	20/3/2026
2.2	User interface (UI) and interaction flow design (student and teacher modules)	8 Days	23/3/2026	1/4/2026
2.3	AR learning flow and 3D character design	8 Days	2/4/2026	13/4/2026
2.4	Database schema and user management design	6 Days	14/4/2026	21/4/2026
2.5	Gamification, AI Tutor and Teacher Dashboard workflow planning	6 Days	22/4/2026	23/4/2026
3.0	**PHASE 3: DEVELOPMENT**	**113 Days**	**24/4/2026**	**25/8/2026**
3.1	Unity project setup and Vuforia SDK integration	21 Days	24/4/2026	22/5/2026
3.2	Student login and profile module	7 Days	25/5/2026	2/6/2026
3.3	AR learning module (character scanning and information display)	18 Days	3/6/2026	24/6/2026
3.4	KOMSAS quiz module	12 Days	25/6/2026	10/7/2026
3.5	Literature anthology and Bahasa Istana module	12 Days	13/7/2026	28/7/2026
3.6	KOMSAS AI Tutor module	12 Days	29/7/2026	13/8/2026
3.7	Gamification (XP, rewards and badges) and progress tracking	8 Days	14/8/2026	21/8/2026
3.8	Teacher Dashboard module	4 Days	24/8/2026	25/8/2026
4.0	**PHASE 4: TESTING**	**18 Days**	**26/8/2026**	**16/9/2026**
4.1	Unit testing for individual modules	6 Days	26/8/2026	2/9/2026
4.2	Integration testing (AR, quiz, gamification and database)	6 Days	3/9/2026	10/9/2026
4.3	User acceptance testing with students and teachers	6 Days	11/9/2026	16/9/2026
5.0	**PHASE 5: DEPLOYMENT**	**16 Days**	**17/9/2026**	**6/10/2026**
5.1	Functional testing on Android devices	6 Days	17/9/2026	24/9/2026
5.2	Usability and user experience testing	5 Days	25/9/2026	1/10/2026
5.3	Bug fixing, optimization and final build (APK)	5 Days	2/10/2026	6/10/2026
6.0	**PHASE 6: REVIEW**	**20 Days**	**7/10/2026**	**21/10/2026**
6.1	Final report writing and documentation	10 Days	7/10/2026	14/10/2026
6.2	Presentation slides and project demonstration preparation	6 Days	15/10/2026	20/10/2026
6.3	Final verification, deployment and submission	2 Days	21/10/2026	21/10/2026

2.0 REQUIREMENT SPECIFICATION

2.1 FUNCTIONAL REQUIREMENT

The functional requirements of the project are as follows:

User Requirements

i. Students shall be able to scan KOMSAS character markers using AR technology

ii. Students shall be able to answer KOMSAS-related quizzes

iii. Students shall be able to use the KOMSAS AI Tutor

iv. Students shall be able to view their learning progress, rewards and badges

v. Teachers shall be able to monitor student learning progress

vi. Teachers shall be able to create quizzes and learning missions

vii. Teachers shall be able to provide rewards to students

System Requirements

i. The system shall authenticate users before providing access to the application

ii. The system shall detect registered KOMSAS markers and display the corresponding 3D character

iii. The system shall display KOMSAS character information and provide audio narration

iv. The system shall provide KOMSAS-related quiz questions and calculate quiz scores

v. The system shall provide immediate feedback after students submit quiz answers

vi. The system shall update XP, coins, badges and learning levels based on completed activities

vii. The system shall provide Literature Anthology and Bahasa Istana learning content

viii. The system shall send KOMSAS-related questions to the AI Tutor and display the generated response

ix. The system shall display student learning progress and leaderboard rankings

x. The system shall allow teachers to create quizzes, learning missions and rewards through the Teacher Dashboard

2.2 NON-FUNCTIONAL REQUIREMENT

The non-functional requirements of the project are as follows:

i. **Usability**

The application shall provide a simple and user-friendly interface that allows students and teachers to navigate the system easily.

ii. **Performance**

The application shall provide responsive interactions and minimise unnecessary loading during learning activities.

iii. **Reliability**

The application shall operate consistently during AR scanning, quizzes, learning activities and navigation between screens.

iv. **Compatibility**

The application shall be compatible with supported Android mobile devices and provide a consistent user experience.

v. **Security**

The system shall authenticate users and restrict teacher-specific functions to authorised teacher accounts.

vi. **Maintainability**

The application shall use a modular structure that allows learning content, quizzes, AR characters and system features to be maintained independently.

vii. **Accessibility**

The application shall use readable text, clear layouts and appropriately sized interactive elements for mobile users.

2.3 HARDWARE AND SOFTWARE REQUIREMENTS

Table 2.1: Cost of Hardware

No	Hardware	Specifications	Quantity	Price
1	Acer Nitro V 15	Intel Core i5-13420H, NVIDIA GeForce RTX 4050 6GB, 15.6" Full HD display, 16GB RAM, 512GB SSD	1	RM 3800
2	vivo Y29s 5G	MediaTek Dimensity 6300, 6GB RAM, 128GB storage, 6.74" LCD, Android 15, 5G	1	RM 800
3	Samsung Galaxy Tab A9	8.7" display, 4GB RAM, 64GB storage, Android	1	RM 487
TOTAL : RM 5087				

Table 2.2: Cost of Software

No	Software	Specifications	Quantity	Price
1	Unity 3D	Unity 6 Personal – game/AR application development	1	Free
2	Vuforia Engine	Basic Plan – AR image target tracking	1	Free
3	Firebase	Spark Plan – authentication, Firestore and cloud services	1	Free
4	Meshy AI	Pro – 3D model generation	1	RM 40
5	Mixamo	3D character rigging and animation	1	Free
6	ChatGPT Plus	AI-assisted development, coding and documentation	1	RM 14
7	Claude Pro	AI-assisted coding and project development	2	RM 172
8	Google AI Studio / Gemini API	AI Tutor integration and KOMSAS question answering	1	Free
TOTAL : RM 226				

2.4 SYSTEM CONFIGURATION

The system configuration of the project are as follows:

i. **Install the KOMSAS GO application**

Users install the application by downloading and executing the .APK file on Android devices, ensuring camera permissions are enabled for AR marker scanning and cloud connectivity is established.



Figure 2.1: KOMSAS GO Application Logo

ii. **Log in to KOMSAS GO application**

Users enter their registered email and password to authenticate their identity securely through Firebase, directing them to the appropriate student or teacher dashboard based on their role.



Figure 2.2: KOMSAS GO Login Screen

2.5 SECURITY REQUIREMENT

The security measures of the project are as follows:

i. **Login Authentication**

The system requires users to authenticate using their registered account credentials before accessing the application.

ii. **Role-Based Access Control**

The system separates Student and Teacher access so that teacher-specific functions are only available to authorised teacher accounts.

iii. **Authenticated Data Access**

The system uses authenticated access when retrieving and storing user learning progress and related data through Firebase services.

iv. **Input Validation**

The system validates required user inputs and form data to prevent incomplete or invalid information from being processed.

3.0 FINAL DESIGN

3.1 LOGICAL DESIGN

i. Use Case Diagram

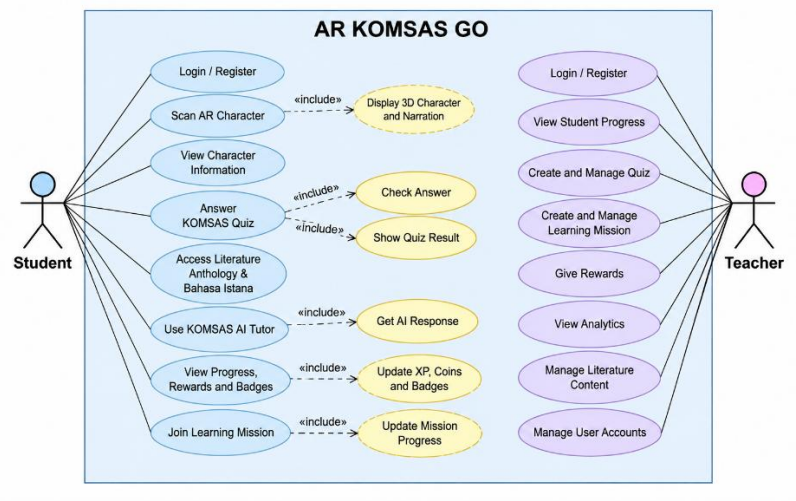


Figure 3.1: AR KOMSAS GO Use Case Diagram

ii. Functional Decomposition Diagram (FDD)

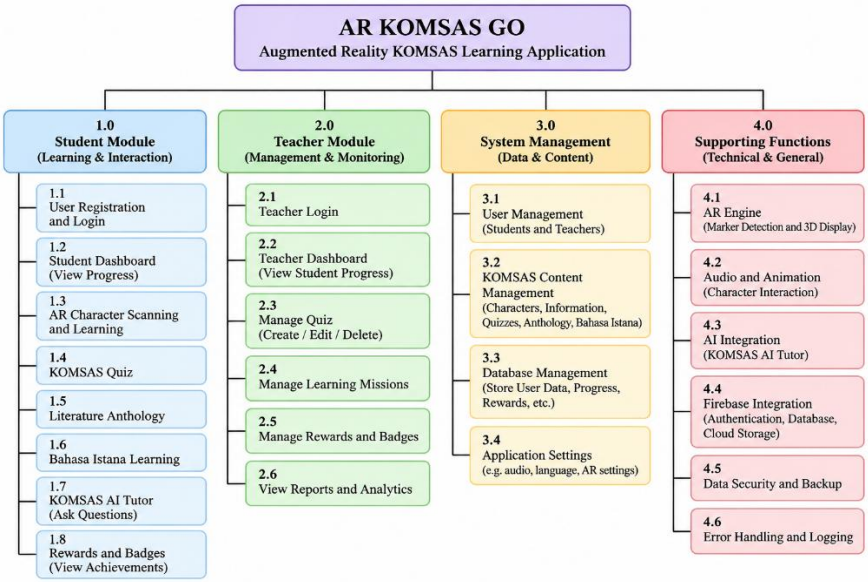


Figure 3.2: AR KOMSAS GO Functional Decomposition Diagram (FDD)

iii. Context Diagram

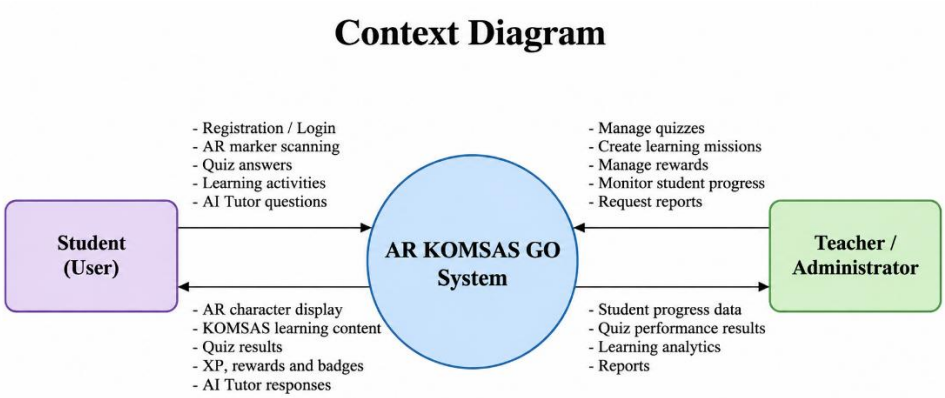


Figure 3.3: AR KOMSAS GO Context Diagram

iv. DFD Diagram

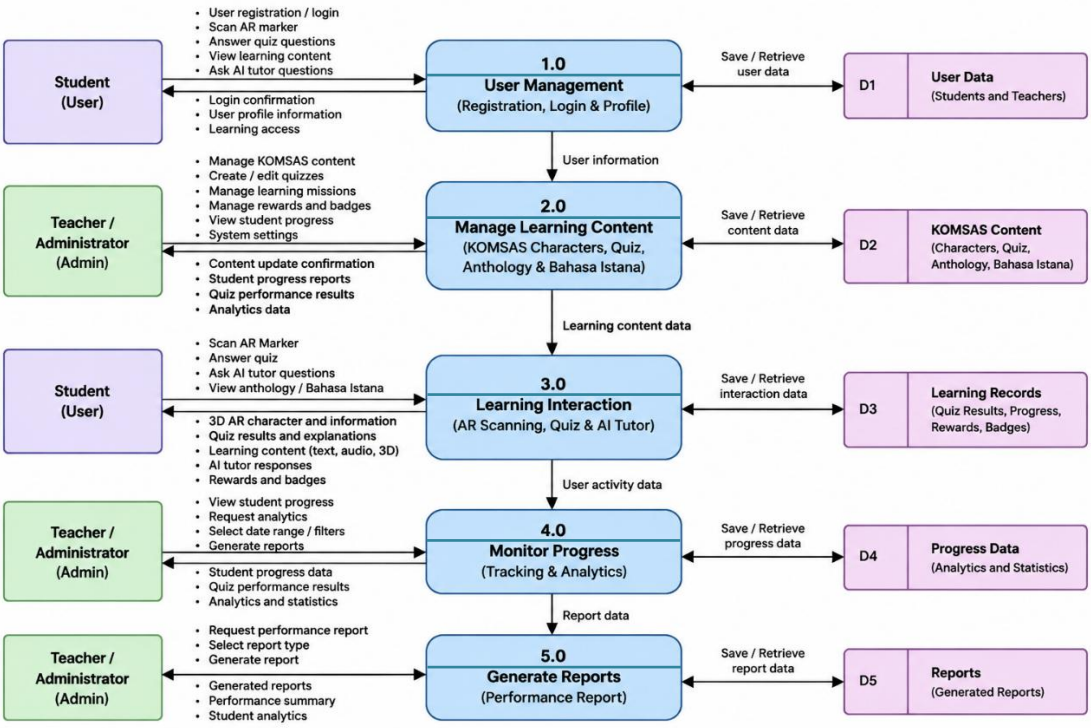


Figure 3.4: AR KOMSAS GO DFD Diagram

3.2 PHYSICAL DESIGN

The physical design of AR KOMSAS GO consists of the user interfaces developed for students and teachers. The interfaces were designed to provide a clear, consistent and user-friendly experience across the main learning and management functions of the application. The following figures present the final interfaces of the developed system:

i. User Login Interface

The application login interface is designed specifically for user authentication, where either students or teachers must enter a valid email address and password to gain full access to the system according to their respective roles, ensuring data security is maintained before beginning the learning session.



Figure 3.5: User Login Interface

ii. Student Dashboard

The main view of the student module that serves as a personal navigation hub. It displays a brief user profile, current level status, XP progress bar, and clear, easily accessible shortcut buttons to quickly initiate AR scanning, answer quizzes, or explore other modules.

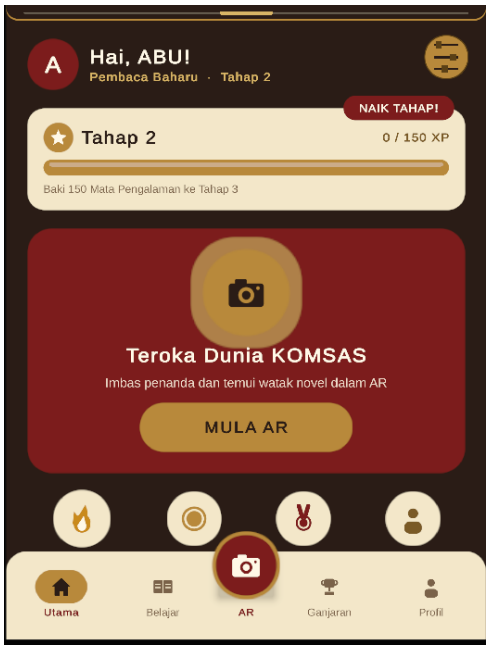


Figure 3.6: Student Dashboard

iii. AR Character Scanning

The application's live camera interface during the physical marker detection process. This view allows the device to detect the correct page to smoothly display KOMSAS novel character models in augmented reality (AR) over real-world surfaces.

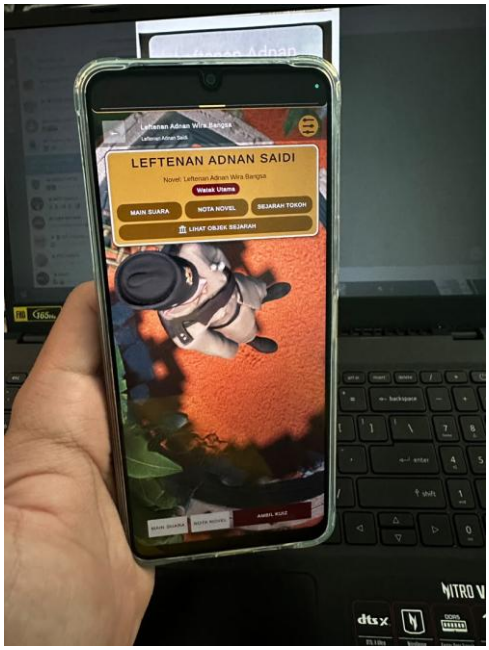


Figure 3.7: AR Character Scanning

iv. AR Character Information

The in-depth information display regarding selected novel characters (such as Leftenan Adnan) that appears simultaneously alongside the interactive 3D model. This screen is equipped with various interaction options such as listening to main voiceover recordings, brief novel notes, and historical figure references.



Figure 3.8: AR Character Information

v. Quiz Interface

The interactive quiz question interface that integrates the application's virtual environment elements to allow students to test their understanding of KOMSAS literary works through multiple-choice question formats answered directly and effectively.

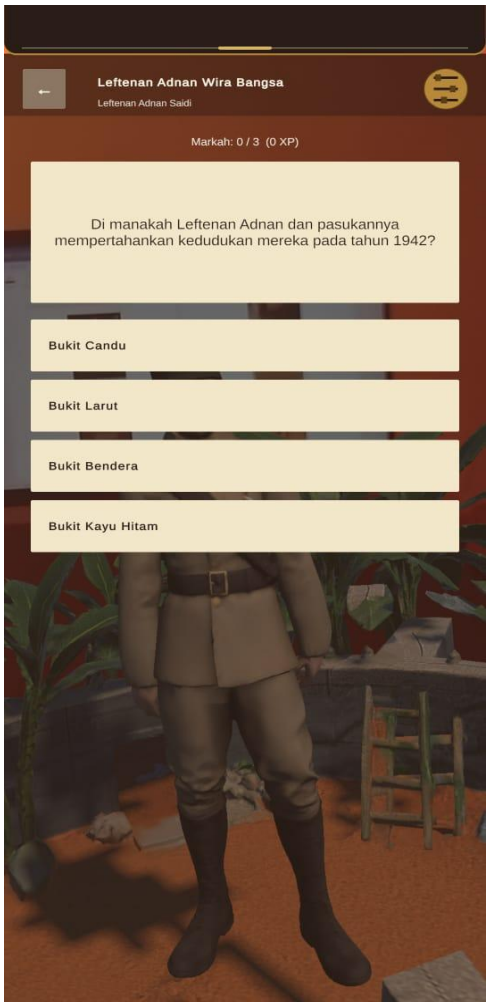


Figure 3.9: Quiz Interface

vi. Gamification Interface

The display of personal gamification elements and badge achievements for students. It shows the status of badges that have been successfully unlocked or remain locked as an additional motivational push to keep students committed to continuing their learning activities.



Figure 3.10: Gamification Interface

vii. Achievement Interface

The comprehensive Rewards interface that displays the total accumulated virtual currency (coins) and total experience points (XP) earned, accompanied by a real-time list of badge achievements that can be unlocked throughout the application's usage period.



Figure 3.11: Achievement Interface

viii. Literature Anthology

The Literature Anthology module directory interface that organizes learning reference materials neatly into several main categories such as Poetry (Puisi), Traditional Prose (Prosa Tradisional), and Royal Court Language (Bahasa Istana) to facilitate quick reference and access by students.



Figure 3.12: Literature Anthology

ix. AI Tutor Interface

The virtual smart assistant interface (KOMSAS AI Tutor) designed to help students interact directly. It allows learners to submit any questions related to character analysis, themes, moral values, or literary concepts to be answered automatically in Malay.



Figure 3.13: AI Tutor Interface

x. Teacher Dashboard

The main dashboard view specifically for teachers that summarizes overall class statistics (including the number of students, daily activity, class average level, and average quiz performance) as well as individual student progress lists for effective monitoring.

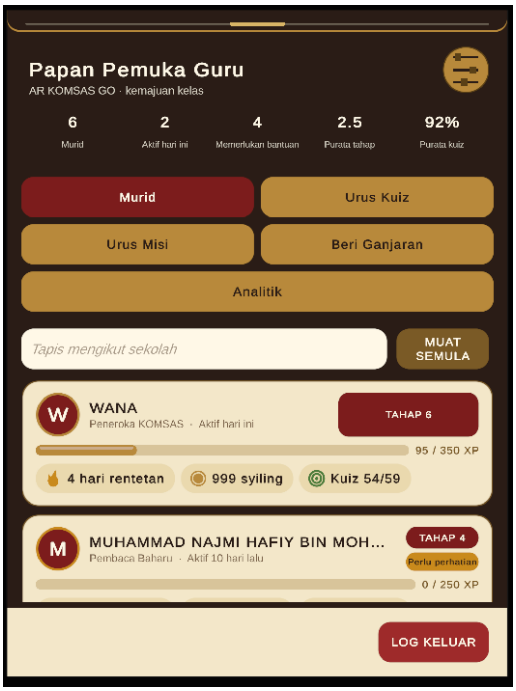


Figure 3.14: Teacher Dashboard

xi. Teacher Mission Management

The learning mission management interface where teachers can create new tasks or missions for students, set specific achievement targets, and offer reward incentives in the form of XP and coins in an organized manner.

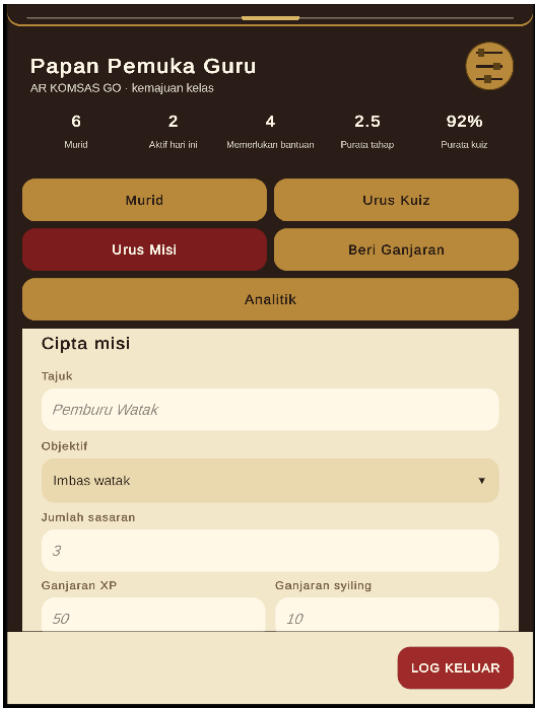


Figure 3.15: Teacher Mission Management

xii. Teacher Quiz Management

The quiz management and editing interface for teachers, allowing them to build, add, update, and delete questions and new quiz answer options easily to suit the learning syllabus needs of students.

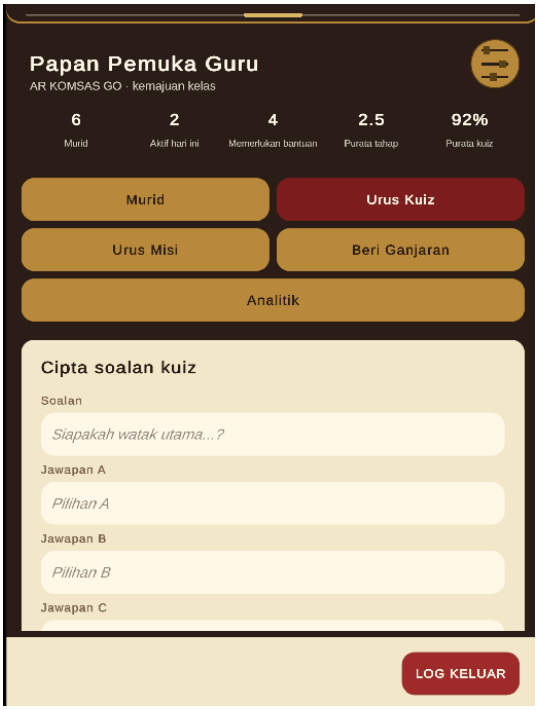


Figure 3.16: Teacher Quiz Management

4.0 TEST DESCRIPTION AND RESULTS

4.1 UNIT TESTING PLAN

Table 4.1: Unit Testing Plan

No	Test Case Name	Test Procedure	Pre-Condition	Expected Result	Tester	Result
1	User Login Validation	Enter valid and invalid login credentials	Login page is displayed	Valid credentials allow login and invalid credentials display an error message	Najmi	Pass
2	Quiz Answer Validation	Select an answer and submit the quiz	Quiz question is available	The selected answer is evaluated correctly	Najmi	Pass
3	Quiz Score Calculation	Complete a quiz with correct and incorrect answers	Quiz is available	The system calculates the correct score	Najmi	Pass
4	Mission Validation	Create a mission with valid and invalid information	Teacher is logged in	Valid mission information is accepted and invalid information is rejected	Hafizuddin	Pass
5	Reward Validation	Select a student and provide a reward	Teacher is logged in	The reward is processed and the correct confirmation message is displayed	Hafizuddin	Pass

6	Literature Content Display	Open the Literature Anthology module	Student is logged in	Literature categories and content are displayed correctly	Adam	Pass
7	Bahasa Istana Display	Open the Bahasa Istana section	Literature Anthology is accessible	The eight Bahasa Istana terms and meanings are displayed correctly	Adam	Pass
8	AI Tutor Input Validation	Submit a KOMSAS-related question to the AI Tutor	Internet connection and AI service are available	The question is accepted and an appropriate response is displayed	Hafizuddin	Pass

4.2 INTEGRATION TESTING PLAN

Table 4.2: Integration Testing Plan

No	Test Case Name	Test Procedure	Pre-Condition	Expected Result	Tester	Result
1	AR Marker and Character Integration	Scan a registered KOMSAS character marker	Camera and AR marker are available	The correct 3D character is displayed after scanning	Najmi	Pass
2	Quiz and Gamification Integration	Complete a quiz and check the student's progress	Student is logged in and quiz is available	Quiz result updates the student's XP and progress correctly	Najmi	Pass
3	Firebase Authentication and Dashboard Integration	Login using a registered account	Firebase connection is available	The correct user dashboard is loaded based on the authenticated account	Adam	Pass
4	Teacher Dashboard and Student Data Integration	Teacher opens student progress information	Teacher is logged in and student data exists	The dashboard displays the correct student progress and statistics	Hafizuddin	Pass
5	Teacher Quiz and Student Quiz Integration	Teacher creates a quiz and student accesses it	Teacher and student accounts are available	The created quiz is available to the student with the correct questions and answers	Hafizuddin	Pass

6	AI Tutor and External API Integration	Submit a KOMSAS question through the AI Tutor	Internet connection and valid API configuration are available	The AI Tutor sends the question and displays the returned response correctly	Najmi	Pass
7	Reward and Student Progress Integration	Teacher gives a reward to a student	Teacher and student accounts are available	The reward is recorded and reflected in the student's account	Adam	Pass
8	Literature Module Navigation Integration	Navigate from the learning dashboard to Literature Anthology and Bahasa Istana	Student is logged in	The user can navigate between the modules without errors	Hafizuddin	Pass

4.3 USER ACCEPTANCE TEST

Table 4.3: User Acceptance Testing

No	Test Case Name	Test Procedure	Pre-Condition	Expected Result	Tester	Result
1	Student Login	Student enters valid login credentials and accesses the application	Student account is registered	Student can successfully access the student dashboard	Mr. Mohmad Nazer	Pass
2	AR Character Learning	Student scans a KOMSAS character marker using the mobile camera	AR marker and application are available	The correct 3D character and information are displayed	Mr. Mohmad Nazer	Pass
3	KOMSAS Quiz	Student completes a KOMSAS quiz and submits the answers	Quiz is available	The quiz score and feedback are displayed correctly	Mr. Mohmad Nazer	Pass
4	Gamification Features	Student completes learning activities and checks XP, rewards and badges	Student account has completed an activity	The student's progress, XP, rewards or badges are updated correctly	Mr. Mohmad Nazer	Pass
5	Literature Learning	Student opens the Literature Anthology and Bahasa Istana module	Student is logged in	The student can access and read the available literature content	Mr. Mohmad Nazer	Pass

6	KOMSAS AI Tutor	Student submits a KOMSAS-related question to the AI Tutor	Internet connection and AI service are available	The AI Tutor provides a relevant response in Bahasa Melayu	Mr. Mohmad Nazer	Pass
7	Teacher Dashboard	Teacher logs in and views student learning progress	Teacher account and student data are available	Teacher can view the relevant student progress and statistics	Mr. Mohmad Nazer	Pass
8	Teacher Content Management	Teacher creates a quiz or learning mission and provides a reward	Teacher is logged in	The created content/reward is successfully processed and available to students	Mr. Mohmad Nazer	Pass

5.0 MAJOR FINDINGS AND DISCUSSION

5.1 THE ADVANTAGES OF THE PROJECT

i. The application integrates Augmented Reality to provide an interactive way for students to explore KOMSAS characters

ii. The application combines learning content, quizzes, gamification and rewards in a single platform

iii. The KOMSAS AI Tutor provides immediate assistance for students with KOMSAS-related questions

iv. The Literature Anthology and Bahasa Istana modules provide additional KOMSAS learning resources in a structured format

v. The Teacher Dashboard allows teachers to monitor student progress, create quizzes and missions, and provide rewards

vi. The gamification features such as XP, badges, levels, missions and leaderboards encourage students to engage with learning activities

5.2 THE DISADVANTAGES OF THE PROJECT

i. The AR features require a compatible Android device with a working camera and suitable lighting conditions

ii. The AI Tutor requires an internet connection and depends on the availability of the external AI service

iii. The AI Tutor may occasionally provide inaccurate or incomplete responses and therefore should be used as a learning aid rather than the sole source of information

iv. The application currently focuses on selected KOMSAS learning content and does not cover all possible Malaysian literature topics

v. Some application features depend on Firebase services and may be affected by network connectivity or service availability

vi. The application is currently designed for Android devices and does not provide native support for other mobile platforms

6.0 RECOMMENDATION AND CONCLUSION

6.1 RECOMMENDATION

Several improvements can be implemented in future versions of AR KOMSAS GO. First, the KOMSAS AI Tutor can be enhanced through a secure backend architecture to protect the API credentials and improve the reliability of AI services. The AI Tutor can also be further developed to provide more structured explanations, examples and personalised learning recommendations based on students' learning progress.

Second, the application can be expanded to cover a wider range of KOMSAS literature content, including additional novels, characters, themes, literary elements and Bahasa Melayu learning topics. More interactive AR characters and learning activities can also be added to provide greater variety for students.

Third, future versions can include more advanced learning analytics in the Teacher Dashboard, such as detailed performance reports, topic-based analysis and identification of students who require additional support. Offline learning capabilities and support for additional mobile platforms can also be considered to improve accessibility.

Finally, a larger-scale user acceptance study involving more students and teachers can be conducted to evaluate the usability, engagement and effectiveness of the application in a real educational environment.

6.2 CONCLUSION

In conclusion, AR KOMSAS GO was successfully developed as a mobile learning application that integrates Augmented Reality, gamification, literature resources, Artificial Intelligence and teacher management features to support KOMSAS learning. The application allows students to explore KOMSAS characters through AR, complete quizzes and missions, earn rewards and badges, access Literature Anthology and Bahasa Istana resources, and obtain learning assistance through the KOMSAS AI Tutor. The application also provides a Teacher Dashboard that enables teachers to monitor student progress, create quizzes and learning missions, and provide rewards. The integration of these features provides a more interactive and engaging learning environment compared with relying solely on conventional learning materials. The development of AR KOMSAS GO using an Agile methodology allowed the system to be developed iteratively, with continuous testing, evaluation and improvement throughout the development process. Although the application has limitations such as dependence on compatible mobile devices, internet connectivity and external AI services, it provides a strong foundation for future development of interactive digital learning tools for KOMSAS education. Overall, AR KOMSAS GO demonstrates the potential of combining Augmented Reality, gamification and Artificial Intelligence in a single educational application to support students and teachers in the learning and teaching of KOMSAS.

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